

Assignment No. 2

1)

⇒ $L = 25\%$ $SP = 450$
 $CP = ?$

$$\begin{aligned} &= \frac{75}{100} = \frac{450}{x} \\ &= 75x = 45000 \\ &= x = 600 \end{aligned}$$

2)

⇒ $CP = 1200$ $SP = 1440$ $P = ?$

$$\begin{aligned} &\frac{1200}{100} = \frac{240}{x} \\ &1200x = 24000 \\ &x = 20\% \end{aligned}$$

$$\begin{array}{r} 1440 \\ - 1200 \\ \hline 240 \end{array}$$

3)

⇒ $SP = 960$ $CP = 800$ $P = ?$

$$\frac{800}{100} = \frac{160}{x}$$

$$= 8x = 160$$

$$= x = 20\%$$

$$\begin{array}{r} 960 \\ 800 \\ \hline 160 \end{array}$$

4)

⇒ $SP = 1200$ $L = 20\%$ $CP = ?$

$$\frac{80}{100} = \frac{1200}{x}$$

$$4x = 6000$$

$$x = 1500$$

5)

⇒

$CP = 400$

$SP = 480$

$P = ?$

$$\frac{400}{100} = \frac{80}{x}$$

$$4x = 80$$

$$x = 20\%$$

6)

discount 20% and 10%

net discount ?

⇒

$$100 \xrightarrow{-20} 80 \xrightarrow{-10} 72$$

$$80 \times \frac{10}{100} = 8$$

$$\left. \begin{array}{r} 80 \\ - 8 \\ \hline 72 \end{array} \right\}$$

$$72$$

7)

$SP = 800$ discount = 20%

marked price ?

⇒

If suppose $CP = 100$

$$\frac{80}{100} = \frac{800}{x}$$

$$= 4x = 4000$$

$$x = 1000$$

8)

$SP = 1800$

$P = 25\%$

$CP = ?$

⇒

$$\frac{175}{100} = \frac{1800}{x}$$

$$= \frac{5}{4} = \frac{1800}{x}$$

$$= 5x = 7200$$

$$= x = 1440$$

9)

$$= CP = 1500 \quad du = 10\%$$

$$1500 \times \frac{10}{100} = 150$$

$$SP = 1500 - 150$$

$$SP = 1350$$

10)

3)

$$\frac{50}{150} \times 100 = 33.33\%$$

11)

=>

15% dis on item make
profit 20%

=>

$$100 \xrightarrow{-15} 85$$

$$x \xrightarrow{+20\%} 120$$

$$\begin{array}{r} 6 \\ +20 \\ \hline 100 \end{array} = \frac{85}{x}$$

$$6x = 425$$

$$x = 70.83$$

$$CP = 70.83$$

$$10 - 70.83 = 29.17 \approx 30$$

$$\frac{30}{70} \times 100 = 42.85\%$$

12)

$$SP = 2250$$

$$P = 10\%$$

$$CP = ?$$

$$\frac{2250 \times 10}{110} = 225$$

$$\left. \begin{array}{r} 2250 \\ 225 \\ \hline 2025 \end{array} \right\}$$

=>

$$x \xrightarrow{+10\%} 225$$

$$\frac{110}{100} = \frac{2250}{x}$$

$$11x = 22500$$

$$x = 2045$$

13)

⇒

$$P = 25\% \quad CP = 800$$

$$SP = ?$$

$$\frac{800 \times 25}{100} = 200$$

$$SP = 800 + 200 = 1000$$

Q14)

⇒

$$SP = 15000 \quad L = 10\%$$

$$CP = ?$$

$$\frac{90}{100} = \frac{15000}{x}$$

$$9x = 1500000$$

$$x = 16666$$

15)

$$CP = 100 \quad \text{then } MP = 150$$

$$SP 20\% = 150 \times 0.80$$

$$= 120$$

$$\text{profit} = \frac{120 - 100}{100} \times 100$$

$$= 20\%$$

16)

$$CP = 400 \quad P = 12\% \quad \text{Dis} = 5\%$$

$$SP 12\% = 400 \times 1.12 = 448$$

$$MP 5\% = \frac{448}{1 - 0.05} = 471$$

17)

⇒

$$CP = 480$$

$$SP = 576$$

$$P = 96$$

$$P = \frac{96}{480} \times 100$$

$$= 20\%$$

18)

$$P = 50$$

$$CP = 500$$

$$P\% = \frac{50}{500} \times 100$$

$$P = 10\%$$

19)

$$P = 15\%$$

$$CP = 2300$$

$$CP = \frac{2300}{1.15} = 2000$$

20)

$$L = 20\%$$

$$SP = 640$$

$$CP = \frac{640}{0.80}$$

$$= 800$$

20)

$$CP = 750$$

$$SP = 900$$

$$P = 900 - 750$$

$$P = 150$$

$$P = \frac{150}{750} \times 100$$

$$= 20\%$$

22)

⇒

$$P = 20\%$$

$$SP = 9600$$

$$CP = \frac{9600}{1.20}$$

$$CP = 8000$$

23)

$$P = 20\%$$

$$SP = 500$$

$$CP = \frac{500}{1.20}$$

$$= 416.67$$

24)

$$\text{Total } CP = 1500 + 1500 = 3000$$

$$SP1(20\% P) = 1500 \times 1.20$$
$$= 1800$$

$$SP2(10\% L) = 1500 \times 0.90$$
$$= 1350$$

$$\text{Total } SP = 1800 + 1350$$
$$= 3150$$

$$\text{Net profit} = 3150 - 3000$$
$$= 150$$

$$P\% = \frac{150}{3000} \times 100$$

$$= 5\%$$

25)

⇒

$$L = 12\% \quad SP = 1250$$

$$CP = \frac{1250}{0.88}$$

$$= 1420.45$$

26)

Assume CP of 1 article = 160

Then SP for 0.5 article = 200

$$\text{Total SP for total article} = 200 \times 2 = 400$$

$$P = \frac{400 - 160}{160} \times 100$$

$$P = 300\%$$

27)

let no to be x

$$\text{equ}^n \& 2x(x \times 0.20x)$$

$$= 490$$

$$= 0.40x^2 = 490$$

$$x^2 = 1225$$

$$x = 35$$

28)

⇒

Assume SP = ~~100~~ 50

= 5% of SP

$$SP = \frac{50}{0.05} = 1000$$

$$CP(20\% \text{ loss}) = \frac{1000}{0.80} = 1250$$

$$\begin{aligned}
 \text{Total loss} &= \text{CP} - \text{SP} + \text{selling cost} \\
 &= 1250 - 1000 + 150 \\
 &= 300
 \end{aligned}$$

30)

$$\Rightarrow \text{CP} = 6000, \quad \text{SE} = 50 = 110\% \text{ of loss}$$

$$\text{loss} = \frac{50}{1.10} = 45.45$$

$$\begin{aligned}
 L\% &= \frac{45.45}{6000} \times 100 \\
 &= 0.76\%
 \end{aligned}$$

31)

$$\Rightarrow \text{let CP of 1 article} = 100$$

$$P = 200$$

$$\begin{aligned}
 \text{SP} &= \text{CP} + P \\
 &= 100 + 200 \\
 &= 300
 \end{aligned}$$

$$\begin{aligned}
 P\% &= \frac{200}{100} \times 100 \\
 &= 200\%
 \end{aligned}$$

32)

$$\Rightarrow \text{let CP} = x$$

$$\begin{aligned}
 \text{Initial profit} &= 20\% \text{ of CP} \\
 &= 500
 \end{aligned}$$

$$0.20x = 500$$

$$x = 2500$$

$$\begin{aligned}
 \text{Initial SP} &= \text{CP} + P \\
 &= 2500 + 500 \\
 &= 3000
 \end{aligned}$$

$$\begin{aligned} \text{New CP} &= \text{decrease by } 20\% \\ &= 2500 \times 0.80 \\ &= 2000 \end{aligned}$$

$$\begin{aligned} \text{New profit} &= 3000 - 2000 \\ &= 1000 \end{aligned}$$

33)

$$\begin{aligned} \Rightarrow \text{CP} &= 100 \quad \text{initial SP (25\% P)} = 125 \\ \text{New CP (decrease 10\%)} &= 100 \times 0.90 \\ &= 90\% \end{aligned}$$

$$\begin{aligned} \text{New P\%} &= \frac{125 - 90}{90} \times 100 \\ &= 38.8\% \end{aligned}$$

34)

$$\Rightarrow \text{CP} = 100, \quad \text{initial SP 50\% profit} = 600$$

$$\text{New CP} = 100 \times 2 = 200$$

$$\text{New SP} = 600 \div 2 = 300$$

$$\begin{aligned} \text{New P\%} &= \frac{300 - 200}{200} \times 100 \\ &= 50\% \end{aligned}$$

35)

$\Rightarrow$ initial price = 100, new price = 125
To spend some amount, consumption decrease by

$$= \frac{125 - 100}{125} \times 100$$

$$= 20\%$$

36)
⇒

$$\begin{aligned}
 CP &= 100 & CP \text{ of } 15 &= 1500 \\
 P &= CP \text{ of } 2 \text{ article} = 200 \\
 \text{profit} &= \frac{200 \times 100}{1500} \\
 &= 13.33\%
 \end{aligned}$$

37)
⇒

$$\begin{aligned}
 0.40a &= 0.50b \\
 \frac{a}{b} &= \frac{0.50}{0.40} = \frac{5}{4}
 \end{aligned}$$

38)

⇒

$$\begin{aligned}
 \text{let discount} &= x \text{ then} \\
 MP &= 5x \\
 &= 5x - x = 4x \\
 x &= 4 \\
 &\text{4 times discount}
 \end{aligned}$$

39)

⇒

$$x = 20\% \text{ of } 12\% \text{ of } 120\% \text{ of } 6250$$

$$\begin{aligned}
 x &= 0.25 \times 0.12 \times 1.20 \times 6250 \\
 &= 0.20 \times 0.12 \times 7500 \\
 x &= 180
 \end{aligned}$$

40)

⇒

$$\begin{aligned}
 CP &= 500 \\
 P &= 100\% \text{ of } SP = 1000 \\
 MP &= \frac{1000}{0.65} = 1538.46
 \end{aligned}$$

41)

⇒

$$\begin{aligned} \text{let } P &= 100 \text{ then } A = 125 \\ B \text{ is smaller than } A \text{ by} \\ &= \frac{125 - 100}{125} \times 100 \\ &= 20\% \end{aligned}$$

42)

⇒

$$SP = CP$$

$$\text{let } CP = x \text{ then dis} = x$$

$$MP = SP + \text{dis}$$

$$= 10000 + x + x$$

$$x = 3333.33$$

$$CP = SP = 3333.33$$

43)

=

$$CP = 70\% \text{ of } SP$$

$$\text{dis} = 40\% \text{ of } SP$$

$$MP = SP + \text{dis}$$

$$= 12600 = SP + 0.40 SP$$

$$= 9000$$

$$CP = 70\% \text{ of } SP$$

$$= 6300$$

44)

⇒

$$\text{let no be } x$$

$$\frac{1}{3}x = 20 + \frac{1}{6}x$$

$$\frac{1}{3}x - \frac{1}{6}x = 20$$

$$x = 120$$

$$120\% \text{ of } x = 1.20 \times 120$$

$$= 144$$

45)

let x

$$0.20x = 20 + 0.20 \times 20$$

$$0.20x = 20 + 4$$

$$x = 120$$

46)

$\Rightarrow$

let x

after doubling & tripling twice

$$x \times 2 \times 3 \times 2 \times 3$$

$$36x$$

$$\text{Percentage change} = \frac{36x - x}{x} \times 100$$

$$= 3500\%$$

47)

$\Rightarrow$

$$65\% \text{ of } 234$$

$$0.65 \times 234$$

$$= 152.1$$

$$\text{reduction needed} = 234 - 152.1$$

$$= 81.9$$

48)

$=$

$$0.90 \times 9 \times 90 \times 9$$

$$= 0.90 \times 9 \times 810$$

$$= 0.90 \times 7290$$

$$= 6561$$

49)

49)

⇒

$$\begin{aligned}
 &\text{Let salary per emp} = 100 \\
 &\text{total initial expenditure} = 2500 \\
 &\text{after layoff} = 12 \text{ emp} \\
 &\text{new salary} = 129 \\
 &\text{new expenditure} = 12 \times 129 \\
 &\quad = 1488 \\
 &\text{decrease \%} = \frac{2500 - 1488}{2500} \times 100 \\
 &\quad = 40.48\%
 \end{aligned}$$

50)

⇒

$$\begin{aligned}
 &\text{CP} = 3500 \\
 &\text{discount} = 15\% \\
 &\quad = 0.15 \times 3500 \\
 &\quad = 525
 \end{aligned}$$